

Package ‘ofhsyn’

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Type Package

Title Synthetic Linked Health Cohort Data Generator

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Author Hannah Nicholls [aut, cre]

Maintainer Hannah Nicholls <hannahnicholls@qmul.ac.uk>

Description Generates synthetic linked health datasets for method development, including participant, questionnaire, clinic measurements, outpatient, inpatient, emergency, mortality, primary care medication, and geography outputs. Supports reproducible generation with configurable cohort size and user-defined ICD-10, OPCS-4, and BNF code pools.

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Encoding UTF-8

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Depends R (>= 4.2.0)

Imports methods,
utils,
stats

Suggests testthat (>= 3.0.0),
knitr,
rmarkdown

VignetteBuilder knitr

Config/testthat/edition 3

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generate_ofh_cohort	<i>Generate Synthetic OFH Cohort Datasets</i>
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Description

Generate linked synthetic health datasets for a configurable cohort.

Usage

```
generate_ofh_cohort(
  n = 5000,
  seed = 42,
  icd10 = NULL,
  icd10_file = NULL,
  opcs4 = NULL,
  opcs4_file = NULL,
  bnf_codes = NULL,
  bnf_codes_file = NULL,
  proportions = NULL,
  record_multipliers = NULL,
  code_config = NULL,
  save_csv = TRUE,
  return_objects = TRUE,
  output_dir = NULL
)
```

Arguments

n	Total synthetic cohort size.
seed	Random seed.
icd10	Optional named character vector of ICD-10 descriptions.
icd10_file	Optional path to a TXT/CSV file containing ICD-10 code and description pairs. TXT format should be tab-separated with code and description columns. CSV format should provide code and description columns.
opcs4	Optional named character vector of OPCS-4 descriptions.
opcs4_file	Optional path to a TXT/CSV file containing OPCS-4 code and description pairs. TXT format should be tab-separated with code and description columns. CSV format should provide code and description columns.
bnf_codes	Optional BNF input for primary care meds. Can be either a character vector of BNF codes or a data frame with columns for code, name, and formulation (optional strength).
bnf_codes_file	Optional path to a TXT/CSV file for BNF input. TXT supports one BNF code per line. CSV supports either code-only or structured medication rows containing code, name, and formulation (optional strength).
proportions	Optional named list of dataset-level coverage proportions. Names should match names(ofh_default_proportions()).
record_multipliers	Optional named list of multipliers for multi-record datasets. Names should match names(ofh_default_record_multipliers()).

code_config	Optional nested list overriding field-level code generation probabilities and pools. Structure should follow ofh_default_code_config().
save_csv	Whether to write CSV outputs to disk.
return_objects	Whether to return generated data frames as an R object.
output_dir	Output directory when save_csv = TRUE.

Value

Named list of generated data frames when return_objects = TRUE; otherwise invisible NULL.

Acknowledgement

We extend our thanks to GitHub user [@icallumwebb](#) for contributing a bug fix that improved custom code handling.

Examples

```
out <- generate_ofh_cohort(n = 200, seed = 123, save_csv = FALSE, return_objects = TRUE)
names(out)
```

generation_primitives *Standalone Synthetic Generation Primitives*

Description

Utility functions for generating participant populations and event-level synthetic records.

Usage

```
generate_ofh_population(n = 1000, seed = 123)

add_inpatient_events(
  data,
  events_per_person = 5,
  icd10_codes = c("I210", "I500", "I639", "E110", "J440"),
  opcs4_codes = c("K401", "K451", "K561", "M011", "E033"),
  seed = 123
)

synthesize_drug_exposure(
  data,
  drug_list = c("0212000B0", "0601023A0"),
  seed = 123,
  mean_items_per_person = 2
)
```

Arguments

data	Input data frame containing a pid column.
n	Number of participants.
seed	Random seed.
events_per_person	Mean events per participant.
icd10_codes	ICD-10 code pool.
opcs4_codes	OPCS-4 code pool.
drug_list	Medication code pool.
mean_items_per_person	Mean prescription items per participant.

Value

Data frame output for each function.

OFHCohortSynthesizer *Reference Class for OFH Cohort Generation*

Description

Reference class API for configuring and running synthetic cohort generation.

Usage

```
OFHCohortSynthesizer
```

Details

Create an instance with `OFHCohortSynthesizer$new(...)` and run generation via `$run_all(n = ...)`.

Examples

```
syn <- OFHCohortSynthesizer$new(project_root = ".", seed = 123)
out <- syn$run_all(n = 100, save_csv = FALSE, return_objects = TRUE)
```

ofh_build_config*Configuration Helpers for OFH Generation*

Description

Helper functions that return default settings and compose full generation configuration lists.

Usage

```
ofh_default_proportions()
ofh_default_record_multipliers()
ofh_default_code_config()
ofh_build_config(
  n = 5000,
  proportions = ofh_default_proportions(),
  record_multipliers = ofh_default_record_multipliers(),
  code_config = list()
)
```

Arguments

n	Total cohort size.
proportions	Dataset proportions list.
record_multipliers	Record multiplier list for event datasets.
code_config	Optional code configuration overrides.

Value

A named list configuration object.

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